

EEG x Deep Learning: what is in the signal, and what the networks learn from it

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All numbers come from `python main.py` (fixed seeds, runs in the CSVs named in brackets). Choices are justified in `DECISIONS.md` (D-numbers) and predictions are scored in `PREDICTIONS.md`.

Task 1: Motor imagery, subjects 70-109

A1 Data audit

Runs 4, 8 and 12 are imagined left vs right fist (T1 = left, T2 = right), verified in the `mne.datasets.eegbci` docstring and on the PhysioNet page (D1). Subjects 88, 92 and 100 carry a 128 Hz header that is correct (no 0.8x alpha shift, consistent protocol), so they are kept and resampled to 160 Hz (D2). Twenty trials whose -1.5 to 4.0 s window overruns the recording are dropped (D3), leaving 897 left and 896 right trials (`audit.csv`).

A2 Physiological ground truth

The reference is undocumented and a lateral one would bias C3 and C4 unequally, so I use a local Laplacian (C3 or C4 minus the mean of its 4 nearest electrodes, D4, D10), whose zero-sum weights cancel it. Morlet power relative to a pre-cue baseline (-1.0 to -0.1 s, D5) is averaged over 0.5-4.0 s, and $LI = (ipsi - contra) / (|ipsi| + |contra|)$ per band is tested with a one-sided label permutation (10 000 permutations, Bonferroni over μ and β), counting only a contralateral desynchronisation (D6). **14 of 40 subjects show lateralised ERD** (`a2_lateralisation.csv`, mean LI 0.26 μ , 0.32 β), and CAR and raw referencing give smaller LIs (0.19/0.29 and 0.15/0.21).

A3 Confound audit

The cue is a target on one side of the screen, so each non-motor route got a restricted decoder (D12-D17) with a pre-cue twin as a leak check. The chance threshold is an exact one-sided binomial at 5% (63-67% within subject, 52.0% pooled over 1793 trials).

decoder	within-subject mean	subjects above threshold (of 40)	pooled cross-subject	pre-cue twin
F1 frontal, 0-0.5 s (eye)	66.4%	25	71.9%	48.2%
F2 frontal, 0.5-4 s	63.5%	18	75.3%	-
O1 occipital, 0-0.5 s (visual)	56.5%	14	59.3%	49.7%
G1 temporal 30-40 Hz (muscle)	50.8%	1	49.6%	49.3%
M1 sensorimotor μ + β (motor)	54.9%	8	54.6%	52.2%*
R2 previous 1-3 labels (no EEG)	70.8%	31	72.6%	-

*Marginal (permutation p 0.02-0.03 among 10 uncorrected pre-cue twins), not interpreted.

The prediction (motor 56-62%, order at chance) was wrong. The eye route dominates, labels alternate ($z = -17.6$), and a zero-phase filter leaked into the pre-cue window until a causal filter removed it (D18). Per-subject results are in `a3_per_subject_table.csv`.

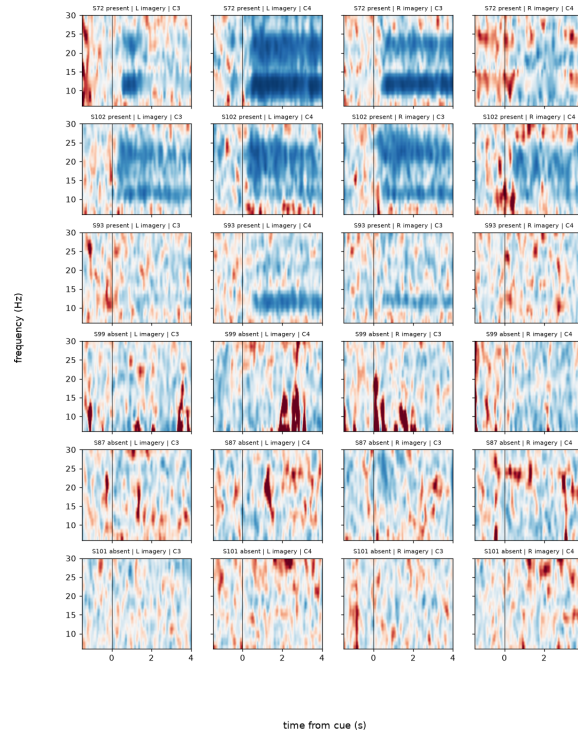


Figure 1: Laplacian-referenced time-frequency power relative to baseline at C3 and C4, for left and right imagery (blue = desynchronisation). Top three rows: lateralised ERD present (S72, S102, S93). Bottom three: absent (S99, S87, S101).

B1 Models and evaluation

Evaluation is cross-subject only (8 fixed subject-wise folds of 5, fold $f+1$ for early stopping, D22). B2 tests on folds 0-3 (20 subjects, 913 trials, threshold 52.8%), seeds 0-2 (D24). EEGNet-8,2 (2962 parameters) has both 0.5 s kernels rescaled from 128 to 160 Hz (64 to 80 and 16 to 20 samples, D20). The transformer (d_model 16, 2 layers, 2 heads) shares EEGNet’s temporal filter and uses time-patch tokens (6834 parameters) or channel tokens with (7250) or without (6226) a learned electrode identity (D21).

model	5 subj.	10	20	30	30, no eye route
EEGNet	72.1 +- 0.5	75.0 +- 1.3	76.2 +- 1.3	76.9 +- 1.0	50.5 +- 2.9
Transformer, time-patch	54.8 +- 3.1	64.3 +- 3.0	66.8 +- 4.1	71.6 +- 3.2	49.1 +- 1.3
Transformer, channel + id				70.2 +- 1.2	
Transformer, channel, no id				57.5 +- 1.6	

Test accuracy %, mean +- SD over 3 seeds (outputs/partb/b2_all_runs.csv). The other B2 experiments are in outputs/partb/B2_RESULTS.md.

Write-up

(a) If a model scores 70% on these subjects, how much of that needs motor imagery?

Almost none. The subject looks at the cue target, and a frontal decoder on the first 0.5 s after the cue (the saccade window, before ERD develops) reaches 71.9% across subjects and passes threshold in 25 of 40, while its pre-cue twin is at chance (48.2%). The sensorimotor decoder reaches only 54.6% and 8/40, although A2 finds lateralised ERD in 14/40, so the motor signal is real but inconsistent across subjects while the eye signal has the same sign in everyone.

The networks follow the eye route. EEGNet (76.9%) and the time-patch transformer (71.6%) rely most on F7, F8, Ft7, Ft8 and Af8 (horizontal EOG positions), and both fall to chance (50.5%, 49.1%) once that route is removed (Laplacian, then the 21 Fc/C/Cp electrodes, D25). The problem is not that they cannot fit motor imagery, it is that what they fit does not transfer, as validation-subject loss rises from the first epoch (D27). A 70% score is therefore fully explainable without motor imagery, with the eye route alone at 72% and the motor route under 55%.

(b) Mechanisms behind the B2 results

Scaling contradicted the predicted crossover. EEGNet leads at every size (72.1% at 5 subjects, 76.9% at 30) and the transformer closes the gap from ~17 to ~5 points without crossing. EEGNet hard-codes the problem as a filter bank with per-band spatial filters, so ~2900 weights pin down the frontal eye pattern from five subjects, while the transformer must learn how 20 time segments combine and that freedom costs data. A crossover needs a signal richer than EEGNet's form, and a spatially broad eye field is not one.

Tokenisation was partly wrong. Time-patch (71.6%) and channel tokens with identity (70.2%) tie, but without identity accuracy falls to 57.5%, because attention with a CLS read-out is then permutation-invariant over electrodes (predictions 100% unchanged under shuffling and left/right mirroring) and ERD under C3 and under C4 look the same.

Displacement ranked as predicted (time-patch drops most), but costs under one point at 10 mm and at most 1.5 at 20 mm. EEGNet's temporal pooling averages the spatial-layer perturbation back from 17% to 13%, while the transformer's grows to 17% because its spatial filters feed every token and the softmax amplifies a coherent change.

Faithfulness ranked as predicted. Deleting the most-attended half of the tokens drops accuracy from 71.4% to 57.7%, against 64.1% for gradient x input and 65.6% for random, because deletion renormalises the softmax, which last-layer CLS attention captures and a first-order gradient estimate does not.

The link to Part A was half right and holds the largest contradiction. Per test subject, EEGNet accuracy correlates with both the A2 LI (Spearman 0.72) and the A3 frontal decoder (0.53), which are unrelated (-0.02), so the network uses both routes in different subjects. I predicted a small drop without the eye route, but it falls to chance because the motor pattern is subject-specific and only the eye pattern is shared.

The intervention (random 0-15 mm cap slides in training) had no effect, as predicted, because to first order it penalises the loss change along a slide, which displacement showed is negligible (accuracy 77.7% vs 76.9%, filter sensitivity 0.40 vs 0.41).

A cross-subject network here learns motor imagery only if the shared non-motor route is removed and the motor pattern transfers across subjects, and on these 40 subjects the second condition fails.

Task 2: Sleep staging pipeline

2a Baseline

The given script, unchanged with seed 42 (task2/baseline/2a_baseline_seed42.txt), scores **accuracy 0.847, macro-F1 0.628, kappa 0.693** (N1 F1 0.207), with 70% of test windows Wake.

2b Defects

Ten defects, one commit each (DEFECTS.md, printouts in task2/evidence/, decisions T2-1 to T2-10).

#	defect	evidence	fix
1	split by recording	5/5 test subjects' other night in training (seed 42)	split by subject, 9/3/3
2	epoch chosen on test	test F1 0.58-0.67 over 12 epochs, max reported	select on validation
3	batch_first=False	own neighbour changes output by 0, other window 9e-4	batch_first=True
4	position after attention	shuffled neighbours change output by 3e-7	encode before
5	label = last epoch	labels differ in 8.7% of windows, 71% of N1	label centre epoch
6	500 "uV" on volts	0 of 80070 epochs rejected, units-only fix rejects 54% (EOG)	uV, EEG only
7	rejected epochs dropped	windows span time gaps	mask, skip windows
8	per-epoch z-score	99.6% of N3 epochs >75 uV, all std = 1 after	per-recording z
9	unscored as Wake	905 "?" + 35 movement epochs became Wake	label -1, context only
10	~23 h recordings	68% of epochs lights-on Wake, "always Wake" = 68.7%	crop to sleep ± 30 min

Fixed pipeline, mean ± SD over seeds 42-44: **accuracy 0.827 ± 0.012, macro-F1 0.788 ± 0.015, kappa 0.769 ± 0.020**.

2c Impact ledger

Each defect is re-introduced alone into the fixed script (`task2/ledger.py`) on seeds 42-44, and Δ is defective minus fixed on the same seed (mean $\pm$ SD, `task2/ledger_table.md`, † = seeds 42 and 43 only).

defect	Δ accuracy	Δ macro-F1	Δ kappa	predicted (F1)	outcome
10 no crop	+0.098 $\pm$ 0.017	+0.013 $\pm$ 0.018	+0.089 $\pm$ 0.028	acc up, F1 $\sim$ 0	right
3 batch attention	-0.026 $\pm$ 0.022	-0.034 $\pm$ 0.020	-0.035 $\pm$ 0.027	down	right (N1 -0.089)
8 per-epoch z	-0.012 $\pm$ 0.010	-0.025 $\pm$ 0.019	-0.020 $\pm$ 0.013	down, N3 most	N1 most
9 unscored as W	-0.015 $\pm$ 0.007	-0.017 $\pm$ 0.007	-0.020 $\pm$ 0.006	$\sim$ 0	larger
2 select on test	+0.010 $\pm$ 0.009	+0.016 $\pm$ 0.014	+0.015 $\pm$ 0.013	up	right
1 recording split	-0.007 $\pm$ 0.023	-0.012 $\pm$ 0.008	-0.014 $\pm$ 0.036	up	wrong sign
4 position after	-0.003 $\pm$ 0.021	-0.006 $\pm$ 0.023	-0.003 $\pm$ 0.027	small down	noise
5 last-epoch label	+0.009 $\pm$ 0.011	+0.007 $\pm$ 0.015	+0.013 $\pm$ 0.014	down	noise
6 units (no rejection)	+0.006 $\pm$ 0.010	+0.008 $\pm$ 0.014	+0.009 $\pm$ 0.016	$\sim$ 0	right (noise)
7 deletion	+0.011 $\pm$ 0.019	+0.008 $\pm$ 0.015	+0.014 $\pm$ 0.024	$\sim$ 0	right (noise)
6N naive units fix †	-0.028 $\pm$ 0.004	-0.049 $\pm$ 0.002	-0.069 $\pm$ 0.000	down	right
6N + 7 †	-0.016 $\pm$ 0.015	-0.021 $\pm$ 0.026	-0.029 $\pm$ 0.024	worse than 6N	flips, less bad
3 + 4	-0.037 $\pm$ 0.016	-0.046 $\pm$ 0.031	-0.048 $\pm$ 0.024	= 3	right
1 + 2	+0.015 $\pm$ 0.033	+0.012 $\pm$ 0.026	+0.014 $\pm$ 0.053	more than either	not additive
9 + 10 †	+0.097 $\pm$ 0.012	+0.005 $\pm$ 0.017	+0.081 $\pm$ 0.021	9 grows	9 vanishes
all ten †	+0.027 $\pm$ 0.031	-0.147 $\pm$ 0.032	-0.059 $\pm$ 0.054	acc up, F1 -0.08..-0.20	right

Noise floor. Runs are bit-identical for the same code, seed and thread count, yet changing 11 of $\sim$ 17,000 training windows (one recording on seed 44) moves macro-F1 by +0.024, so 3 seeds cannot resolve effects under about $\pm$ 0.02 macro-F1.

Ranked by how much each would have misled the report, defect 10 is first, adding $\sim$ 10 points of accuracy and 9 of kappa from lights-on Wake. Defect 3 is second because the context model never ran yet the CNN still scores well, and 8 third because it removes the amplitude criterion of N3 and N1. Defects 9 and 2 are consistent but small, and 1, 4, 5, 6 and 7 sit in the noise floor.

Flags. Defect 4 vanishes under 3, and 7 vanishes under 6 but *flips* with the naive units fix, because EOG rejection of waking eye movements leaves masking 11k training windows against 17.3k for deletion (Wake 2694 to 381). Defect 9 vanishes once Wake is uncropped (10). Defect 1 has the wrong sign because the seed also draws a harder set of test subjects, and with 3 test subjects who is tested matters more than leakage.

2d Physiology: the worst stage

N1 is the worst stage (F1 0.511 $\pm$ 0.017, others 0.83-0.89), with recall 0.61 but precision 0.44, because the model uses it for epochs it cannot place. On EEG and EOG, N1 overlaps REM (median Fpz-Cz amplitude 78 vs 79 μ V), and only muscle tone separates them (EMG 2.68 vs 0.69 μ V), which the pipeline never uses. The expected confusion is N3 scored as N2 (17.5% of N3), because the 20% slow-wave rule thresholds a continuum. The unexpected one is W with REM (5.1% and 3.3%), which human scorers separate by chin atonia (EMG 3.23 vs 0.69 μ V), a channel the pipeline drops. The full analysis is in `task2/PHYSIOLOGY.md`.

2e Note for a non-technical reader

Our first sleep-staging program reported 85% accuracy. The corrected version scores about 83%, and the lower figure is the honest one.

The first test was too easy. Each recording runs for almost a whole day, and for two-thirds of it the person is awake. A program that always answered “awake” would score 69%. The corrected test keeps only the night, plus half an hour either side.

The old test had two other flaws. It included people the program had already studied, since their other night was in its training material. And its best version was picked using the test itself. Real patients will be new to it.

A fairer score gives every sleep stage equal weight, so easy wakefulness earns nothing extra. On that score the corrected program improved from 0.63 to 0.79 (out of 1). The headline dropped because the test got harder, while the program got better at telling sleep stages apart.